TensorFlow – Complete Concept

1. What is TensorFlow?

TensorFlow is an open-source Machine Learning (ML) and Deep Learning (DL) library developed by Google.

It is used to:

- Build AI models
- Train neural networks
- Process images, text, audio, and video
- Create chatbots
- Develop recommendation systems
- Build computer vision applications

Installation

```
pip install tensorflow
```

Import TensorFlow

```
import tensorflow as tf
print(tf.__version__)
```

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2. Why TensorFlow?

Advantages

✅ Easy Deep Learning Development

✅ GPU Support

✅ Production Ready

✅ Mobile Deployment

✅ Large Community

✅ Works with Keras

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3. Tensor

A Tensor is the basic data structure in TensorFlow.

Think of it as a multidimensional array.

Scalar (0D)

```
a = tf.constant(10)
print(a)
```

Output:

10

Vector (1D)

```
a = tf.constant([1,2,3])
```

Output:

```
[1 2 3]
```

Matrix (2D)

```
a = tf.constant([
    [1,2],
    [3,4]
])
```

3D Tensor

```
a = tf.constant([
    [[1,2],[3,4]],
    [[5,6],[7,8]]
])
```

---

4. TensorFlow Data Types

tf.int32
tf.int64
tf.float32
tf.float64
tf.bool
tf.string

Example:

```
a = tf.constant([1,2,3], dtype=tf.float32)
```

---

5. Tensor Operations

Addition

```
a = tf.constant([1,2,3])
b = tf.constant([4,5,6])

print(a+b)
```

Output:

```
[5 7 9]
```

Subtraction

```
print(a-b)
```

Multiplication

```
print(a*b)
```

Division

```
print(a/b)
```

---

6. Tensor Shape

a = tf.constant([[1,2],[3,4]])

print(a.shape)

Output:

(2,2)

---

7. Reshaping Tensor

a = tf.constant([1,2,3,4,5,6])

b = tf.reshape(a,(2,3))

print(b)

Output:

```
[[1 2 3]
 [4 5 6]]
```

---

8. Variables

Variables can change during training.

x = tf.Variable(5)

x.assign(10)

print(x)

---

9. Random Tensors

tf.random.normal((3,3))

tf.random.uniform((3,3))

---

10. TensorFlow Math Functions

tf.reduce_sum()
tf.reduce_mean()
tf.reduce_max()
tf.reduce_min()
tf.sqrt()
tf.square()

Example:

a = tf.constant([1,2,3,4])

print(tf.reduce_sum(a))

Output:

10

---

11. Gradient Tape

Used for automatic differentiation.

```
x = tf.Variable(3.0)

with tf.GradientTape() as tape:
    y = x**2

dy_dx = tape.gradient(y,x)

print(dy_dx)
```

Output:

```
6.0
```

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12. Keras in TensorFlow

Keras is TensorFlow's high-level API.

```
from tensorflow import keras
```

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13. Building First Neural Network

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(10, activation='relu'),
```

```
    Dense(1)
])
```

---

14. Model Compilation

```
model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['accuracy']
)
```

Optimizer

- Adam
- SGD
- RMSprop
- Adagrad

Loss Functions

- MSE
- MAE
- Binary Crossentropy
- Categorical Crossentropy

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15. Training Model

```
model.fit(X_train,
          y_train,
          epochs=10)
```

---

16. Prediction

pred = model.predict(X_test)

---

17. Evaluation

model.evaluate(X_test,y_test)

---

18. Saving Model

model.save("model.h5")

Load:

from tensorflow.keras.models import load_model

model = load_model("model.h5")

---

19. Layers in TensorFlow

Dense Layer

Dense(128)

Dropout Layer

Prevents overfitting.

Dropout(0.5)

Flatten Layer

Flatten()

BatchNormalization

BatchNormalization()

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20. Activation Functions

ReLU

relu

Most commonly used.

Sigmoid

sigmoid

Binary classification.

Softmax

softmax

Multi-class classification.

Tanh

tanh

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21. CNN (Convolutional Neural Network)

Used for image processing.

Layers:

Conv2D
MaxPooling2D
Flatten
Dense

Example:

```
model = Sequential([
    Conv2D(32,(3,3),activation='relu'),
    MaxPooling2D(),
    Flatten(),
    Dense(10,activation='softmax')
])
```

Applications:

- Face Detection
- Object Detection
- Medical Imaging

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22. RNN (Recurrent Neural Network)

Used for sequence data.

Examples:

- Text
- Speech
- Time Series

Layers:

SimpleRNN
LSTM
GRU

Example:

LSTM(128)

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23. LSTM

Long Short-Term Memory Network.

Used for:

- NLP
- Chatbots
- Stock Prediction
- Language Translation

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24. Transfer Learning

Using a pre-trained model.

Popular Models:

- VGG16
- ResNet
- MobileNet
- EfficientNet

Example:

base_model = tf.keras.applications.MobileNetV2()

---

25. TensorFlow Dataset API

dataset = tf.data.Dataset.from_tensor_slices(data)

Benefits:

- Fast Loading
- Efficient Training
- Parallel Processing

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26. TensorBoard

Visualization tool.

tensorboard --logdir logs

Used for:

- Graph Visualization
- Loss Tracking
- Accuracy Monitoring

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27. GPU Support

Check GPU:

print(tf.config.list_physical_devices('GPU'))

---

28. TensorFlow Lite

TensorFlow for Mobile Devices.

Used in:

- Android Apps
- IoT Devices
- Edge AI

converter = tf.lite.TFLiteConverter.from_saved_model(path)

29. TensorFlow Serving

Deploy trained models.

Features:

- REST API
- Production Deployment
- High Performance

30. TensorFlow Project Workflow

Collect Data
↓
Clean Data
↓
Preprocess Data
↓
Build Model
↓
Compile Model
↓
Train Model
↓
Evaluate Model
↓
Save Model
↓
Deploy Model

TensorFlow Interview Questions

Basic

1. What is TensorFlow?
2. What is a Tensor?
3. Difference between Tensor and Variable?
4. What is Keras?
5. What is Gradient Tape?

Intermediate

6. What is CNN?
7. What is RNN?
8. What is LSTM?
9. What is Transfer Learning?
10. What is TensorBoard?

Advanced

11. Explain Backpropagation.
12. Explain Adam Optimizer.
13. Difference between CNN and RNN.
14. How does Batch Normalization work?
15. How does TensorFlow use GPUs?

TensorFlow Learning Roadmap

Phase 1: Basics

- Tensor

- Variables
- Operations
- Tensor Shapes
- Reshape

Phase 2: Keras

- Sequential Model
- Functional API
- Layers
- Activation Functions

Phase 3: Deep Learning

- ANN
- CNN
- RNN
- LSTM

Phase 4: Advanced

- Transfer Learning
- TensorBoard
- GPU Training
- TensorFlow Lite
- TensorFlow Serving

Phase 5: Projects

- House Price Prediction
- Customer Churn Prediction
- Handwritten Digit Recognition
- Face Mask Detection
- Sentiment Analysis
- Chatbot
- Image Classification

Master these topics and you'll be ready for most TensorFlow interviews, projects, and practical ML development.